

Capital Structure, Dividends, and Share Repurchases

Shaping a modern corporation's financial profile might appear to be an infinitely complex task. But in practice, it typically boils down to just three decisions: how much to invest, how much debt to carry, and how much cash to return to shareholders. In this book, we devote most of our attention to exploring the first of these topics, but the others are also important. It's not so much that making the right decisions about capital structure will create a great deal of value; it's that making the wrong calls can destroy tremendous amounts of it. For example, during the high-tech bubble of the late 1990s, many European telecommunication companies accumulated unprecedented levels of debt on their balance sheets to fund investments in digital mobile networks, expecting to issue equity at a later stage to repay the borrowing. But before they could, the bursting of the high-tech bubble in 2000 drove down the earnings outlook for mobile services and the share prices for telecom players. Providers had to recapitalize their balance sheets at great pain and cost, losing billions of shareholder value.

The primary objective of a company's decisions to structure its capital, pay dividends, and repurchase shares should be to ensure that the company has enough capital to pursue its strategic objectives and to weather any cash shortfalls along the way. If a company doesn't have enough capital, it will either pass up opportunities or, worse, fall into financial distress or even bankruptcy. When a company holds too much capital, the remedy is much easier: it can always increase its cash distributions to shareholders.

This chapter explores the options managers have for choosing an appropriate capital structure for their company and how they should develop a supporting policy for returning cash to shareholders or raising new capital. In the first two sections, we discuss some practical guidelines and a four-step

approach to deciding a company's capital structure, payout, and financing. The remainder of the chapter discusses key theoretical and empirical findings on capital structure and payout that form the basis for our guidelines and approach.

PRACTICAL GUIDELINES

Finance theory has much to say about capital structure and payout—for example, about the costs and benefits of leverage, the way markets react to shareholder payouts, and the ability of managers to time their buying back of shares.¹ But it does not tell us how to set an effective capital structure and payout policy for a given company. Building on insights from finance theory (explored later in this chapter), we offer the following practical guidelines to help executives make the right choices on capital structure and payout:

- *Decisions about capital structure, dividends, and share repurchases should be an integral part of overall cash deployment.* This matches investment needs across businesses with funding opportunities and payouts to shareholders to best support the company's strategy and risk preferences. When deciding to deploy cash (for example, by using it for share repurchases), companies should consider all alternative uses of cash and set priorities for the uses according to their potential to create value, as laid out in Exhibit 33.1. The greatest opportunity to create value comes from investing cash in business operations (organic growth) and acquisitions at returns above the cost of capital.² The returns are typically higher for organic growth, making it the first choice for deploying cash. One level below is using cash for growth by acquisitions, where returns on capital tend to be somewhat lower because acquiring assets usually requires paying a premium.³ Financing—that is, using (or raising) cash to adjust a company's capital structure—should assume a lower priority. This does not mean that capital structure decisions are unimportant; rather, they are a necessary means of ensuring that sufficient funding is available to capture attractive investment opportunities and withstand cash shortfalls. At the bottom of the list of cash alternatives are payout decisions. These don't drive value directly but should aim to return cash to shareholders when a company has insufficient opportunities to reinvest at returns above the cost of capital.

¹ For an overview of the literature, see M. Barclay and C. Smith, "The Capital Structure Puzzle: The Evidence Revisited," *Journal of Applied Corporate Finance* 17, no. 1 (2005): 8–17.

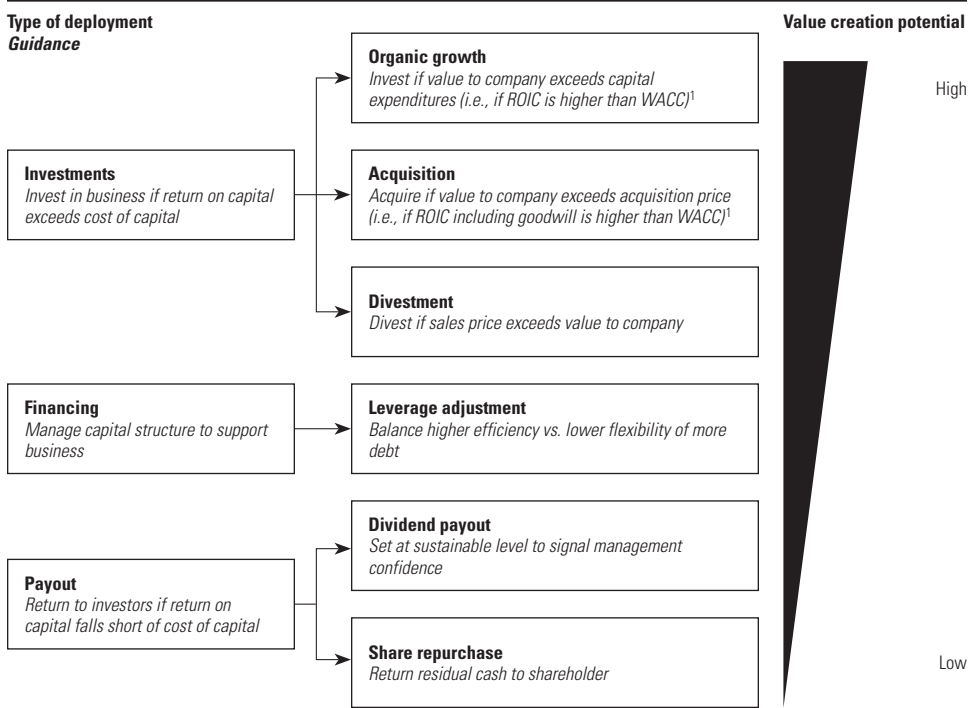
² Following the conservation of value principle in Chapter 4, this is the primary source of value creation for companies.

³ See M. Goedhart and T. Koller, "The Value Premium of Organic Growth," *McKinsey on Finance*, no. 61 (2017): 14–15.

- *For their capital structure, large companies should target investment-grade credit ratings between A+ and BBB– to maintain adequate flexibility for difficult times.* Most large exchange-listed companies worldwide have capital structures in this range of credit ratings. Lower ratings typically lead to a significant loss of flexibility, due to restrictive covenants built into loan agreements for sub-investment-grade companies. Higher credit ratings offer little or no additional benefits, as a company typically has enough flexibility to pursue investment opportunities once it reaches a solid investment-grade rating.
- *Payout decisions should consider their short-term impact on stock prices.* Dividends and share repurchases are value neutral over the long term but can lead to earlier recognition of value creation in a company's share price. Although long-term value creation comes from business operations and investments that generate returns above the cost of capital, not from a company's payouts to shareholders, short-term price increases can result from increased payouts that signal management discipline in the use of capital and confidence in the company's outlook. In applying this guideline, keep in mind that such increases in share price reflect higher expectations of future value creation. If the company fails to meet these expectations, the price will drop again.
- *Dividends should be set at a level that a company can sustain under plausible adverse conditions—for example, during the bottom of the earnings cycle.* Most shareholders expect that regular dividends (or dividend payout ratios) will be cut from customary levels only in cases of severe setbacks.⁴ Investors almost always perceive the cutting of regular dividends as a signal of significantly lower future value creation, so these cuts generally lead to sharp declines in share price and increases in share price volatility.
- *Share repurchases should be used to return excess cash over and above dividend levels to shareholders.* Investors do not consider share buybacks to be the same long-term commitment as regular dividends. As a result, repurchases are a flexible way to pay out cash amounts that vary from year to year. Unlike dividends, share repurchases typically increase a company's earnings per share, but that does not mean share repurchases create value. Keep in mind that repurchasing shares, like paying regular dividends, is value neutral. In fact, both types of payouts could even indirectly destroy value if they come at the expense of attractive investments; that is why these decisions need to be part of planning a company's broader cash deployment.

⁴ A small number of companies have a variable dividend policy that targets a fixed payout ratio (or range) of dividends relative to earnings.

Figure 33.1 **Cash Deployment: Value Creation Hierarchy**



¹ ROIC is return on invested capital; WACC is weighted average cost of capital.

A FOUR-STEP APPROACH

With these guidelines in mind, we recommend a sequential approach to establishing capital structure and payout policies. With a clearly defined corporate strategy in place, the approach itself consists of four stages:

1. Project and stress-test the operating cash flows.
2. Develop a capital structure target based on the company's risk profile and risk appetite.
3. Estimate the surplus or deficit cash flow to shareholders by combining the operating cash flow and the capital structure target.
4. Decide on the payout of cash flow surplus and financing of cash flow deficit, including tactical measures, such as share repurchases, dividend payouts, share issuances, and measures to adjust the company's debt to the specified target levels.

To illustrate the approach, we can apply it to a hypothetical company in international branded consumer products. In the past, the company, which we

call MaxNV, has generated annual operating earnings before interest, taxes, depreciation, and amortization (EBITDA) of around \$1 billion, with some fluctuations resulting from movements in raw-materials prices and currency rates. MaxNV has held little debt, but acquisitions have driven up its ratio of net debt to EBITDA from 1.5 in 2015 to 2.8 at the beginning of 2020 (calculated as net debt at beginning of year over expected EBITDA for the year, which for 2020 would equal \$2.8 billion divided by \$1.0 billion).

Step 1: Project and Stress-Test Operating Cash Flows

MaxNV's strategic plan under a base-case scenario foresees annual EBITDA growth of 5 percent, from \$1.0 billion in 2020 to \$1.2 billion in 2024 (see Exhibit 33.2). Growth derives in part from planned bolt-on acquisitions of around \$0.2 billion per year, with some revenue lost to minor divestments. In the base case, MaxNV generates around \$3.0 billion in free cash flow from operations over the next five years.

We tested some of the most important business risks for MaxNV's key market and product segments by developing two downside scenarios. In a competitive-disruption scenario, new entrants with direct-to-customer sales will be more successful than anticipated. Beyond 2021, this will start to depress price and volume levels and require MaxNV to accelerate acquisitions

EXHIBIT 33.2 **MaxNV: Projections of Operating Cash Flows**

		Projections					Cumulative, 2020–2024
		2020	2021	2022	2023	2024	
EBITDA¹	Base	1,000	1,050	1,103	1,158	1,216	5,526
	Competitive-disruption impact	–	–	(100)	(200)	(200)	(500)
	Economic-downturn impact	–	(100)	(150)	(100)	(100)	(450)
Capital expenditures	Base	(100)	(105)	(110)	(116)	(122)	(553)
	Competitive-disruption impact	–	(50)	(50)	(50)	(50)	(200)
Acquisitions	Base	(200)	(200)	(200)	(200)	(200)	(1,000)
	Competitive-disruption impact	–	–	(500)	–	–	(500)
Divestments	Base	–	25	50	–	–	75
	Competitive-disruption impact	–	–	–	25	25	50
Operating taxes	Base	(188)	(197)	(207)	(217)	(228)	(1,036)
	Competitive-disruption impact	–	–	25	50	50	125
FCF² from operations	Base	513	573	636	625	666	3,012
	Competitive-disruption impact	–	(50)	(625)	(175)	(175)	(1,025)
	Economic-downturn impact	–	(100)	(150)	(100)	(100)	(450)

¹ Earnings before interest, taxes, depreciation, and amortization.

² Future cash flows.

and investments in its own direct-to-customer channels. Compared with the base case, annual EBITDA will be around \$200 million lower and capital expenditures around \$50 million higher by 2024. Including an additional \$500 million spent on acquisitions, MaxNV will generate about \$1.0 billion less in after-tax cash flow from operations than in the base case. The second downside scenario sees this competitive disruption aggravated by a major economic downturn, depressing revenues and earnings across the sector. EBITDA will now be \$300 million lower in 2024 compared with the base case.

For companies in industries where price and volume risks are greater, such as commodities, you might replace the use of scenarios with a more sophisticated approach: modeling future cash flows by using stochastic simulation techniques to estimate the probability of financial distress at the various debt levels.

Step 2: Develop a Capital Structure Target

Next, we set a target credit rating and estimated the corresponding coverage ratios to develop a capital structure target. Although MaxNV's operating performance is normally stable (as it is with most branded-consumer-goods players), we targeted the high end of a BBB credit rating because of the company's currency risk as an exporter. We translated the target credit rating to a target net-debt-to-EBITDA coverage ratio of 2.5 times.⁵ This coverage ratio was applied in all scenarios.

Step 3: Estimate Surplus or Deficit

Based on the target coverage ratio and projections of operating cash flows, we estimated MaxNV's target capital structure and cash surpluses (or deficits) for each of the next five years. The detailed calculations are shown in Exhibit 33.3. For example, in the base case scenario, \$1.0 billion of EBITDA in 2020 and a target coverage ratio of 2.5 times result in a target debt level of \$2.5 billion for the end of the year. Starting with \$2.8 billion of debt at the beginning of 2020, deducting \$513 million of free cash flow from operations and adding \$105 million of after-tax interest expenses leave MaxNV with surplus cash of \$108 million that could be distributed to shareholders in 2020. With the same calculation through the remaining years of the forecast, the cumulative cash surplus for distribution amounts to around \$2.7 billion over the five-year period. Exhibit 33.3 also shows the cumulative surplus for the competitive-disruption scenario (\$1.2 billion) and the economic-downturn scenario (\$552 million).

⁵ As discussed later in this chapter, empirical analysis shows that approximate credit ratings can be estimated well with three factors: industry, size, and interest coverage.

EXHIBIT 33.3 **MaxNV: Estimates of Cash Surplus and Deficit**

\$ million

Base case scenario	Projections					Cumulative, 2020–2024
	2020	2021	2022	2023	2024	
EBITDA ¹	1,000	1,050	1,103	1,158	1,216	5,526
Net debt, beginning of year	(2,800)	(2,500)	(2,625)	(2,756)	(2,894)	(2,800)
FCF ² from operations	513	573	636	625	666	3,012
Interest, after-tax	(105)	(94)	(98)	(103)	(109)	(509)
Add: Target net debt, end of year @ 2.5× EBITDA ¹	2,500	2,625	2,756	2,894	3,039	3,039
Cash surplus paid out to equity (cash deficit funded with debt)	108	604	668	659	702	2,742
Target net debt EOY @ 2.5× EBITDA	(2,500)	(2,625)	(2,756)	(2,894)	(3,039)	
Excess debt	—	—	—	—	—	
Cash deficit funded with debt	(2,500)	(2,625)	(2,756)	(2,894)	(3,039)	
Competitive-disruption scenario						
EBITDA ¹	1,000	1,050	1,003	958	1,016	5,026
Net debt, beginning of year	(2,800)	(2,500)	(2,625)	(2,713)	(2,394)	(2,800)
FCF ² from operations	513	523	11	450	491	1,987
Interest, after-tax	(105)	(94)	(98)	(102)	(90)	(489)
Add: Target net debt, end of year @ 2.5× EBITDA ²	2,500	2,625	2,506	2,394	2,539	2,539
Cash surplus paid out to equity (cash deficit funded with debt)	108	554	(207)	29	546	1,237
Target net debt, end of year	(2,500)	(2,625)	(2,506)	(2,394)	(2,539)	
Excess debt	—	—	(207)	—	—	
Net debt, end of year	(2,500)	(2,625)	(2,713)	(2,394)	(2,539)	
Economic-downturn scenario						
EBITDA ¹	1,000	950	853	858	916	4,576
Net debt, beginning of year	(2,800)	(2,500)	(2,375)	(2,604)	(2,351)	(2,800)
FCF ² from operations	513	423	(139)	350	391	1,537
Interest, after-tax	(105)	(94)	(89)	(98)	(88)	(474)
Add: Target net debt, end of year @ 2.5× EBITDA ²	2,500	2,375	2,131	2,144	2,289	2,289
Cash surplus paid out to equity (cash deficit funded with debt)	108	204	(472)	(207)	240	552
Target net debt, end of year	(2,500)	(2,375)	(2,131)	(2,144)	(2,289)	
Excess debt	—	—	(472)	(207)	—	
Net debt, end of year	(2,500)	(2,375)	(2,604)	(2,351)	(2,289)	

¹ Earnings before interest, taxes, depreciation, and amortization.² Future cash flows.

For both downside scenarios, a cash deficit occurs in some individual years. For these years, MaxNV could decide to simply exceed target debt levels and return to target levels later. Alternatively, it could build up excess debt capacity in prior years to ensure target debt levels are met in each year. Of course, if a cumulative deficit occurred for the entire planning horizon, MaxNV would need to consider issuing equity or find other financing opportunities, such as additional divestitures or cost savings.

Step 4: Decide on a Surplus Payout and Deficit Financing

The final step is to decide what payout and financing over the ensuing years will move the company to its target capital structure. Consider Exhibit 33.4, which summarizes the cumulative cash flows associated with the four steps for each of the three scenarios. Over the next five years under all scenarios, MaxNV can easily return \$450 million (\$90 million per year) in the form of regular dividends. Taking a less conservative stance, MaxNV could even consider a dividend payout of about \$1 billion (\$200 million per year), which it would need to cut back in the case of a downturn scenario. If the new dividend payout represents an increase from current levels, its announcement would send a strong signal to the stock market that MaxNV is confident about its business outlook and its ability to sustain this dividend level.

EXHIBIT 33.4 **MaxNV: Deciding on Payout**

\$ million

	Cumulative cash flows, 2020–2024				
	Base case	Competitive disruption		Economic downturn	
	Scenario	Disruption impact	Scenario	Downturn impact	Scenario
Step 1					
Project operational cash flows					
EBITDA ¹	5,526	(500)	5,026	(450)	4,576
Capital expenditures	(553)	(200)	(753)		(753)
Acquisitions	(1,000)	(500)	(1,500)		(1,500)
Divestments	75	50	125		125
Operating taxes	(1,036)	125	(911)		(911)
Future cash flow from operations	3,012	(1,025)	1,987	(450)	1,537
Step 2					
Develop capital structure target					
Net debt/EBITDA target	2.5		2.5		2.5
Step 3					
Estimate surplus (deficit)					
Net debt, beginning of year 2020	(2,800)		(2,800)		(2,800)
Future cash flow from operations	3,012		1,987		1,537
Interest, after taxes	(509)		(489)		(474)
Add: Target net debt, end of year 2024 @ 2.5× EBITDA	3,039		2,539		2,289
Cash surplus paid out to equity	2,742		1,237		552
Step 4					
Decide on payout (financing)					
Dividend payout	450		450		450
Share buybacks	2,292		787		102
Cash surplus paid out to equity	2,742		1,237		552
Step 5					
Dividend per year, average	90		90		90
Buyback per year, average	458		157		20

¹ Earnings before interest, taxes, depreciation, and amortization.

Any remaining cash for each of the scenarios could be returned to shareholders over the next several years through share repurchases or extraordinary dividends. The amount based on a conservative \$450 million dividend payout would be almost \$2.3 billion under the base case, about \$800 million under the disruption scenario, and about \$100 million under the downturn scenario. Like a dividend increase, share repurchases and extraordinary dividends signal confidence, but they have the advantage that investors won't see them as a commitment to additional payouts in future years. This gives MaxNV valuable flexibility to change the amount of cash paid out over the next years in accordance with business results and market developments. It might increase its payout, for example, as management becomes more certain that the company will achieve the base-case projection, or it could withhold most of the cash as long as it considers a downturn scenario more likely.

SETTING A TARGET CAPITAL STRUCTURE

Financing instruments vary widely, offering many options, from traditional common equity and straight debt to more exotic instruments, among them convertible preferred equity and convertible and commodity-linked debt. But the essential choice remains between straight debt and common equity. In this balancing act, tilting toward equity gives managers more flexibility to work through unexpected downturns or take advantage of unforeseen opportunities, such as acquisitions. Taking on more debt delivers higher efficiency from tax benefits and enhances management discipline over investment spending.

Empirical research shows that companies actively manage their capital structure around certain leverage boundaries.⁶ They make adjustments to regain their target capital structure after they have missed it for one or two years, rather than immediately after each change in leverage. Continual adjustment would be impractical and costly, due to share price volatility and transaction costs.⁷

Fundamental Debt/Equity Trade-Offs

For decades, academic researchers have sought to learn which debt-to-equity ratio represents the best trade-off between flexibility and efficiency and maximizes value for shareholders. Unfortunately, a clear model remains elusive.⁸

⁶ P. Marsh, "The Choice between Equity and Debt: An Empirical Study," *Journal of Finance* 37, no. 1 (1982): 121–144.

⁷ See, for example, M. Leary and M. Roberts, "Do Firms Rebalance Their Capital Structures?" *Journal of Finance* 60, no. 6 (2005): 2575–2619.

⁸ For an overview, see Barclay and Smith, "The Capital Structure Puzzle."

The most obvious benefit of debt over equity is a reduction in taxes. Interest charges for debt are typically tax deductible; payments to shareholders as dividends and share repurchases are not.⁹ Reducing taxes by replacing equity with debt increases a company's aggregate cash flow and its value.¹⁰ That said, this advantage does not necessarily make 100 percent debt funding the most tax-efficient approach. More debt funding may reduce corporate taxes but could actually lead to higher taxes for investors. In many countries, investors pay higher taxes on interest income than on capital gains from equity holdings. Under these circumstances equity funding could prove more attractive than debt, depending on the relevant tax rates for corporations and investors.¹¹

Private-equity firms have known for decades that debt can also impose investment discipline on managers, according to the free-cash-flow hypothesis.¹² Especially in companies with strong cash flows and few growth opportunities, managers may be tempted to increase corporate spending on perks or investment projects and acquisitions that will boost growth at the expense of value. If share ownership is widely dispersed, it is difficult and costly for shareholders to assess when managers are engaging in such overinvestment. Debt restrains such behavior by forcing the company to pay out free cash flow according to scheduled interest and principal obligations before managers can make any additional investments.

However, higher levels of debt reduce financial flexibility for companies. This can give rise to costs from business erosion and investor conflicts.¹³ Highly leveraged companies have less flexibility to pursue investment opportunities or free up budgets for research and development (R&D), since they need cash available to repay debts on time. These companies typically face covenants in loan agreements that limit their freedom of action. When credit is tight, they may also have limited access to new borrowing, especially if their debt is not investment grade. This was the case during the 2008 financial crisis.

As a result, these companies may miss significant opportunities to create value. They are also more likely to lose customers, employees, and suppliers because of their greater risk of financial distress. For example, suppliers to highly indebted retailers typically demand up-front payment, sometimes

⁹ Interest charges are not always deductible in full. Many countries have "thin capitalization rules" that limit interest deductibility for taxes. For example, as of 2018, corporations in the United States can deduct interest charges only up to 30 percent of EBITDA.

¹⁰ For an overview, see M. Grinblatt and S. Titman, *Financial Markets and Corporate Strategy*, 2nd ed. (New York: McGraw-Hill, 2002), chap. 14; and R. Brealey, S. Myers, and F. Allen, *Principles of Corporate Finance*, 13th ed. (New York: McGraw-Hill, 2019), chap. 18.

¹¹ M. Miller, "Debt and Taxes," *Journal of Finance* 32, no. 2 (1977): 261–275.

¹² M. Jensen, "Agency Costs of Free Cash Flow, Corporate Finance and Takeovers," *American Economic Review* 76, no. 2 (1986): 323–339.

¹³ We prefer the term *business erosion* to the more often used *financial distress* because the associated costs arise very gradually and long before there may be an actual distress event, such as nonperformance on debt.

creating a negative cycle of lower inventories that lead to lower sales, which then leads to more difficulty in meeting debt schedules, and so on. The risk of losing customers is particularly high when the products require long-term service and maintenance. For example, Chrysler and General Motors lost considerable market share to Japanese and European competitors as they faced financial distress during the 2008 credit crisis. Ultimately, such business erosion can even lead to bankruptcy.

Higher leverage may cause additional value destruction as a result of conflicts of interest among debt holders, shareholders, and managers. For example, when companies come close to defaulting on their debt, shareholders will prefer to take out cash or invest it in high-risk opportunities, at debt holders' expense.¹⁴ Of course, debt holders anticipate such conflicts and try to protect themselves with restrictive covenants and other costly measures.

Evidence on Debt/Equity Trade-Offs

Although finance theory is clear about the sources of costs and benefits of leverage, it does not tell us specifically how to measure the best capital structure for a given company. Fortunately, it turns out that capital structure has less impact on value than many practitioners think. In addition, evidence from academic research provides some guidance on leverage profiles for companies, depending on their characteristics, as one would expect from fundamental debt/equity trade-offs.¹⁵

Leverage should be higher for companies with higher returns, lower growth and risk, or larger and more fungible assets. Indeed, the most highly leveraged industries are typically mature and asset intensive (think cement, packaged consumer goods, and utilities). Their stable profits enable high tax savings from interest deductibility, and their low growth calls for strong management discipline, given the likelihood of overinvesting. Because such companies have assets that can serve as collateral and be redeployed after bankruptcy, their expected costs of business erosion are lower. This also explains why airlines can sustain high leverage: in spite of their low returns and high risk, airplanes are easily deployed for use by other airline companies in the event of a bankruptcy.¹⁶ Note that direct bankruptcy costs are relatively small, around 3 percent of a company's market value, before the company becomes distressed.¹⁷

¹⁴ In finance theory, these effects from high leverage are called corporate underinvestment (taking out cash rather than investing at low risk) and asset substitution (exchanging lower-risk assets for higher-risk assets). See, for example, S. Ross, R. Westerfield, J. Jaffe, and B. Jordan, *Corporate Finance*, 12th ed. (New York: McGraw-Hill, 2019), chap. 17.

¹⁵ R. Rajan and L. Zingales, "What Do We Know about Capital Structure? Some Evidence from International Data," *Journal of Finance* 50, no. 5 (1995): 1421–1460.

¹⁶ Specifically, leverage is high when the operating leases of aircraft are taken into account.

¹⁷ See, for example, L. Weiss, "Bankruptcy Resolution: Direct Costs and Violation of Priority of Claims," *Journal of Financial Economics* 27, no. 2 (1990): 285–314.

Leverage should be lower for companies with lower returns, higher growth potential and risk, or highly specific assets and capabilities. This is the case in sectors such as software, biotechnology, and high-tech start-ups. Potential tax savings are small, because their taxable profits are low in the near term. Management needs more financial freedom, because investments are essential to capture future growth. In contrast, the costs of business erosion are high, because these companies would quickly lose valuable growth opportunities, and any remaining assets have very little value to third parties. For the same reasons, companies with more volatile earnings and higher advertising and R&D costs are generally financed with less debt.¹⁸ Leverage also tends to be low for companies producing durable goods, such as machinery and equipment, requiring long-term maintenance and support. The highly specific capabilities of these companies make financial distress costly for their customers.¹⁹

Although some finance textbooks show a high potential tax benefit from higher leverage, the benefit is usually limited for large, investment-grade companies. To illustrate, consider a simple example. Exhibit 33.5 shows how the multiple of enterprise value over earnings before interest, taxes, and amortization (EBITA) for an average company in the S&P 500 would change along with the amount of the company's debt financing, as measured by the EBITA-to-interest coverage ratio. The EBITA multiple is estimated using the basic value driver formula, presented in Chapter 3, and applied using an adjusted-present-value (APV) methodology.²⁰ We assume a long-term ROIC of 14 percent and an unlevered cost of capital of 9 percent—typical scores for a middle-of-the-road S&P 500 company. As the exhibit shows, tax-related benefits from debt do not change enterprise value dramatically, except at very low levels

¹⁸ M. Bradley, G. Jarell, and E. Kim, "On the Existence of an Optimal Capital Structure: Theory and Evidence," *Journal of Finance* 39, no. 3 (1984): 857–878; and M. Long and I. Malitz, "The Investment-Financing Nexus: Some Empirical Evidence," *Midland Corporate Finance Journal* 3, no. 3 (1985): 53–59.

¹⁹ See Barclay and Smith, "The Capital Structure Puzzle"; and S. Titman and R. Wessels, "The Determinants of Capital Structure Choice," *Journal of Finance* 43, no. 1 (1988): 1–19.

²⁰ Applying the APV methodology to the value driver formula and discounting the tax shield on interest at the unlevered cost of equity results in the following formula:

$$\text{Value} = \text{NOPAT} \left(\frac{1 - \frac{g}{\text{ROIC}}}{k_u - g} \right) + \sum_{t=1}^{\infty} \frac{k_D \times T \times D_t}{(1 + k_u)^t}$$

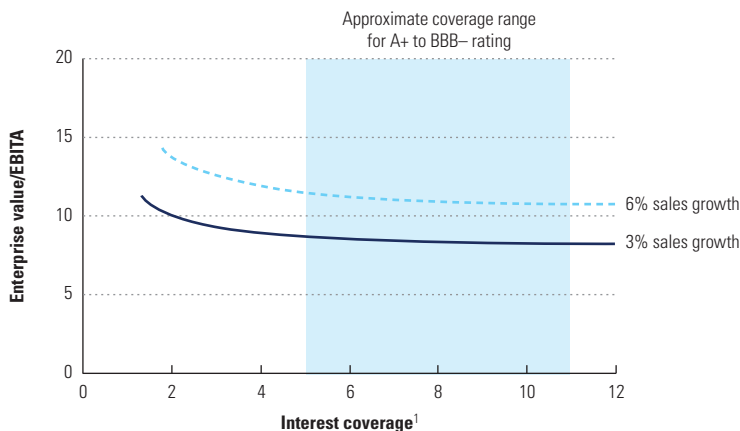
where k_u is the unlevered cost of equity, D_t is the debt in year t , k_D is the cost of debt, T is the tax rate, and all other symbols are as defined in Chapter 3.

If we make the additional assumption that companies finance with debt while maintaining a stable interest coverage ratio, the formula can be simplified as follows:

$$\text{Value} = \text{NOPAT} \left(\frac{1 - \frac{g}{\text{ROIC}} + \frac{T}{1 - T} \left[\frac{\text{Interest}}{\text{EBITA}} \right]}{k_u - g} \right)$$

where EBITA/Interest is the target coverage ratio.

EXHIBIT 33.5 Capital Structure's Limited Impact on Enterprise Value



¹ EBITA/interest.

of interest coverage (below 2) rarely seen for large, investment-grade companies.²¹ Compare that with the much bigger impact on shareholder value of key value drivers such as return on invested capital (ROIC) and growth.

In contrast, losses in flexibility from higher leverage do translate to significant value destruction. John Graham and others examined listed U.S. companies over a period of more than 25 years and analyzed the loss in a company's value due to deviations of its leverage from what was estimated as its theoretical optimum.²² The analysis offers two key insights, illustrated in Exhibit 33.6. First, it confirms our analysis that value at stake is limited to no more than a couple of percentage points for a fairly wide range of leverage around the theoretical optimum. Second, it shows that there is a lot more downside from having too much debt than from having too little. In other words, the losses due to diminished flexibility tend to outweigh the gains from tax benefits and management discipline.

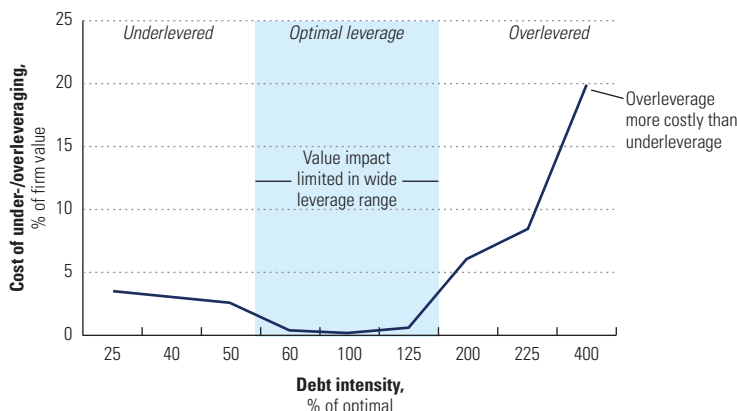
Credit Ratings and Target Capital Structure

Difficult as it may be to determine an *optimal* capital structure, it is much easier to find an *effective* structure—that is, one that cannot clearly be improved upon in terms of shareholder value creation because it is somewhere in the relatively flat range of the valuation curves of Exhibits 33.5 and 33.6.

²¹ Note that at such low levels of coverage, the expected value of any tax savings will itself decline because of the growing probability that the company will not capture these savings in the first place. As a result, the true curve would be even flatter than shown here.

²² See, for example, J. Van Binsbergen, J. Graham, and J. Yang, "The Cost of Debt," *Journal of Finance* 65, no. 6 (2010): 2089–2136.

EXHIBIT 33.6 More to Lose Than to Gain from Capital Structure Management



Source: J. Van Binsbergen, J. Graham, and J. Yang, "The Cost of Debt," *Journal of Finance* 65, no. 6 (2010).

Exhibit 33.7 shows the distribution of credit ratings and the associated average probability of default for all private and public companies worldwide with revenues in 2018 over €1 billion, according to Standard & Poor's. The ratings, which serve as indicators of a company's credit quality, range between AAA (highest quality) and D (defaulted). Ratings of BBB– and higher indicate so-called investment-grade quality. A majority (60 percent) of the companies in Exhibit 33.7 are in the rating categories of A+ to BBB–; an even larger share (71 percent) fall in this range when we consider only companies with a market capitalization over €5 billion. This is apparently an effective rating level: credit ratings are fairly stable over time, so most companies probably do not move in and out of this range. Few companies are at rating levels of AA– and higher, because too little leverage would leave too much value on the table in the form of tax savings and management discipline. At the other extreme, below the rating level of BBB–, the costs of business erosion and investor conflicts associated with high leverage become too onerous. At these ratings, the opportunities for debt funding are also much smaller, because many investors are barred from investing in sub-investment-grade debt.

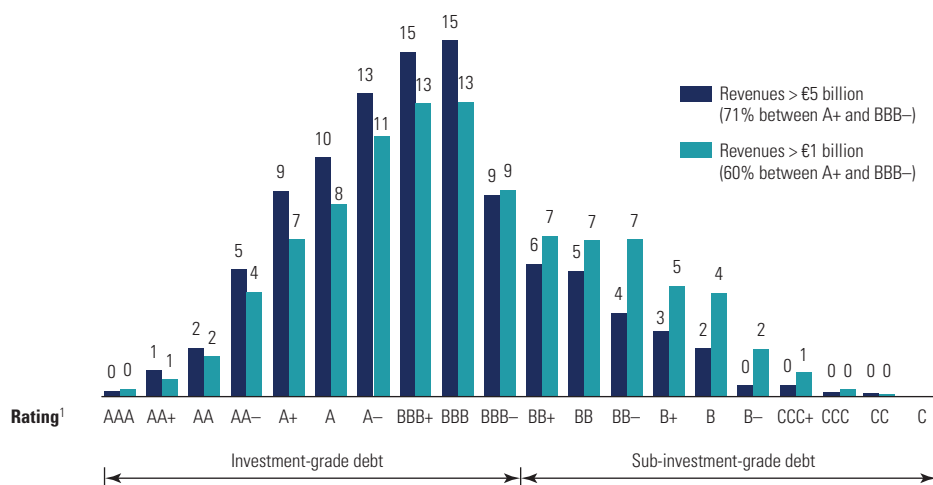
Over the past decade, credit ratings for these large companies have declined on average, shifting the distribution in Exhibit 33.7 to the right.²³ In the United States, some experts and policy makers have expressed concerns about rising levels of corporate debt and deteriorating credit ratings.²⁴ However,

²³ See for example: "Carry the Weight: Should the World Worry about America's Corporate-Debt Mountain?" *The Economist*, March 14, 2019, www.economist.com.

²⁴ See for example, J. Cox, "Yellen and the Fed Are Afraid of a Corporate Debt Bubble, but Investors Still Aren't," CNBC, December 11, 2018, www.cnbc.com; N. Timiraos and A. Ackerman, "Fed Chairman Powell Warns of Economic Risks from Rising Business Debt," *Wall Street Journal*, May 20, 2019, www.wsj.com.

EXHIBIT 33.7 Credit Ratings for Large Companies: Mostly between A+ and BBB–

% of companies sampled, by company size (revenues)



¹ Standard & Poor's credit ratings for all private and public companies with 2018 revenues exceeding €1 billion.

Source: S&P Capital IQ; Corporate Performance Analytics by McKinsey.

these trends do not necessarily mean that companies have taken on too much debt. The fraction of sub-investment-grade companies did indeed increase from 2008 to 2018, but this was driven not so much by downgrades of corporations as it was by newly rated corporations that probably entered debt markets to benefit from historically low interest rates. And while it is true that between 2008 and 2018, corporate debt in the United States grew from \$2.3 trillion to \$5.2 trillion, key credit ratios are still similar to those of the prior ten-year period.²⁵ (Authors' note: As this book went to press in March 2020, companies and governments were just beginning to assess and respond to the economic effects of the global COVID-19 pandemic.)

To translate an investment-grade (AAA to BBB–) rating into a capital structure target for a company, you must understand what a company's credit rating represents and what goes into determining it. Empirical evidence shows that credit ratings are primarily related to two financial indicators.²⁶ The first indicator is *size* in terms of sales or market capitalization. However, this indicator makes a difference only for very large or very small companies. For example, as of 2019, all industrial companies with AAA ratings, such as Microsoft and Johnson & Johnson, have market capitalizations

²⁵ See T. Khurana, W. Rehm, and A. Srivastava, "Is a Leverage Reckoning Coming?" *McKinsey on Finance*, no. 70 (May 2019): 1–6.

²⁶ For an overview, see R. Cantor, "An Introduction to Recent Research on Credit Ratings," *Journal of Banking and Finance* 28, no. 11 (2004): 2565–2573; E. Altman, "Financial Ratios, Discriminant Analysis, and the Prediction of Corporate Bankruptcy," *Journal of Finance* 23, no. 4 (1968): 589–609; and J. Pettit, C. Fitt, S. Orlov, and A. Kalsekar, "The New World of Credit Ratings," UBS research report (September 2004).

above \$350 billion. One possible explanation: larger companies are more likely to diversify their risk.

The second indicator is *coverage* in terms of EBITA or EBITDA relative to interest expense or debt, defined as follows:

$$\begin{aligned}\text{Debt Coverage} &= \frac{\text{Net Debt}}{\text{EBITA}} \text{ or } \frac{\text{Net Debt}}{\text{EBITDA}} \\ \text{Interest Coverage} &= \frac{\text{EBITA}}{\text{Interest}} \text{ or } \frac{\text{EBITDA}}{\text{Interest}}\end{aligned}$$

A similar indicator that is widely used by credit analysts is based on so-called free flow from operations (FFO) instead of EBITA or EBITDA. FFO is defined as EBITDA minus interest and tax charges.

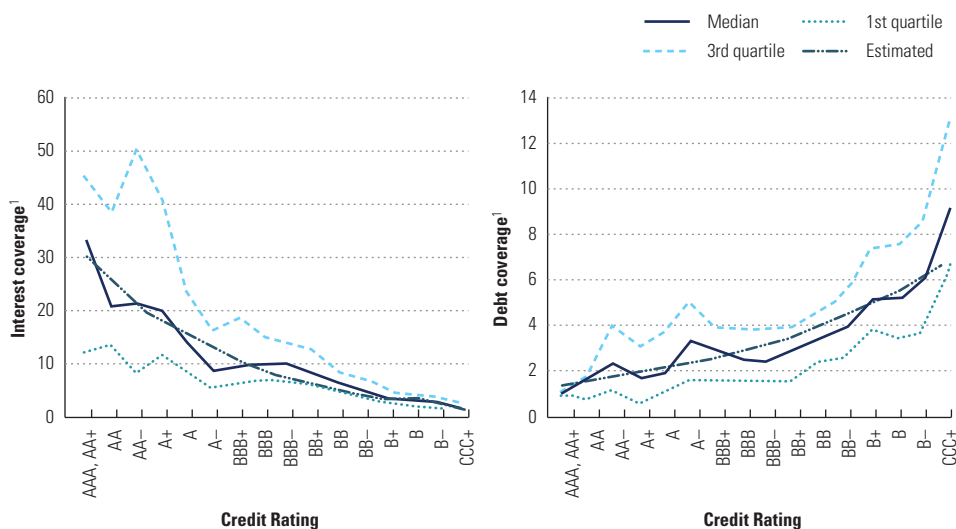
Coverage is more relevant than size when you are setting a capital structure target. Basically, it represents a company's ability to comply with its debt service obligations. For example, EBITA interest coverage measures how many times a company could pay its interest commitments out of its pretax operational cash flow if it invested only an amount equal to its annual depreciation charges to keep the business running (or, for EBITDA coverage, if it invested nothing at all). In today's low-interest-rate environment, however, debt coverage is a better measure of a company's long-term ability to service its debt. Interest coverage ratios might appear strong today for some companies simply because they attracted debt at low interest rates over the past few years. When these companies need to re-fund the debt at higher rates in the future, their interest coverage will plummet.

Exhibit 33.8 shows how interest coverage and debt coverage explain rating differences for a sample of large U.S. companies rated by Standard & Poor's (excluding financial institutions). Obviously, we could further refine the analysis by including more explanatory ratios, such as free flow from operations (FFO) to interest, solvency, and more. However, these ratios are often highly correlated, so calculating them does not always produce a clearer explanation.

For a given credit rating, the coverage will typically differ by industry (see Exhibit 33.9). This is because of differences in underlying business risk. Companies in industries with more volatile earnings need higher coverage to attain a given credit rating, because their cash flow is more likely to fall short of their interest commitments.²⁷ For example, companies in basic materials—say, steel companies—will need higher levels of interest coverage than food and beverage companies to attain the same credit rating. By taking into account these differences in coverage requirements across industries, we can translate a company's targeted credit rating into a target coverage ratio. Based on the company's estimated future operating profit (and interest rate), we can derive

²⁷ Earnings volatility is measured here as the average standard deviation of relative annual changes in EBITDA for companies in each sector.

EXHIBIT 33.8 Credit Rating vs. Interest and Debt Coverage



¹ EBITDA/interest. EBITDA is earnings before interest, taxes, depreciation, and amortization.

² Net debt/EBITDA.

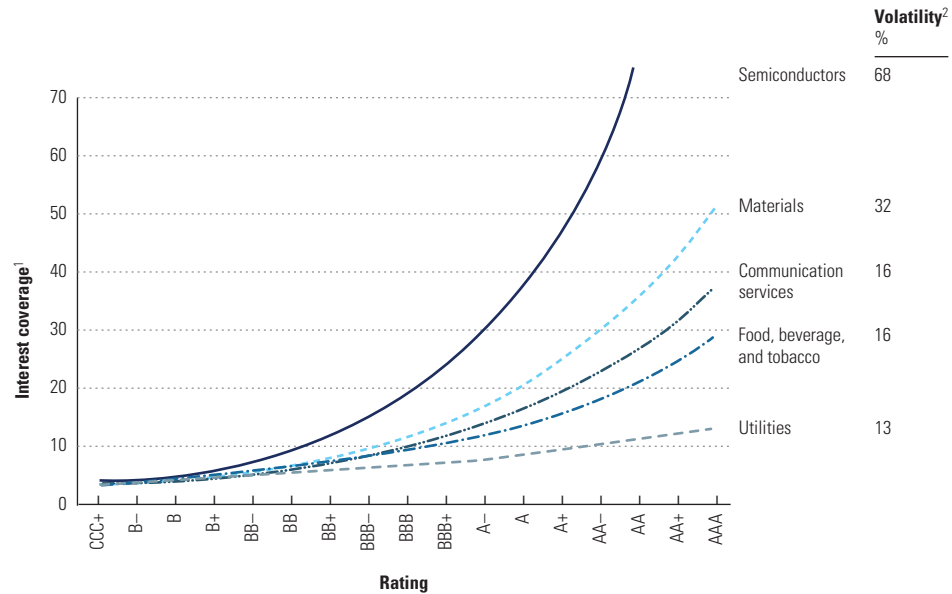
Source: S&P Capital IQ; Corporate Performance Analytics by McKinsey.

its maximum debt capacity for the chosen credit rating and, thereby, its target capital structure. For example, companies aiming for an investment-grade rating in the food and beverage sector would typically need to have an EBITDA-to-interest ratio of around 5 or better. Given projections of near-term EBITDA and interest rates, you can derive a first estimate of the target amount of net debt for such a company to reach an investment-grade rating. A definitive rating estimate would require more in-depth analysis of specific financial and business risks that the company is facing. A place to start is, for example, with the websites for Standard & Poor's (www.spratings.com) or Moody's (www.moody.com).

It is important to compare a target capital structure for a company against that of its industry peer group. The key determinants of value trade-offs in designing capital structure—growth, return, and asset specificity—are largely industry specific, so any large differences in capital structure would require further investigation. It also makes sense from a competitive perspective: as long as your capital structure is not too different, you have at least not given away any competitive advantage derived from capital structure (nor have you gained any).²⁸ Since the 1960s, a body of evidence has built up showing

²⁸ For example, there is academic evidence that high-leverage companies sometimes fall victim to price wars started by financially stronger competitors. See P. Bolton and D. Scharfstein, "A Theory of Predation Based on Agency Problems in Financial Contracting," *American Economic Review* 80, no. 1 (1990): 93–106.

EXHIBIT 33.9 Interest Coverage and Credit Rating for Selected Industry Sectors



¹ EBITDA/interest. EBITDA is earnings before interest, taxes, depreciation, and amortization.

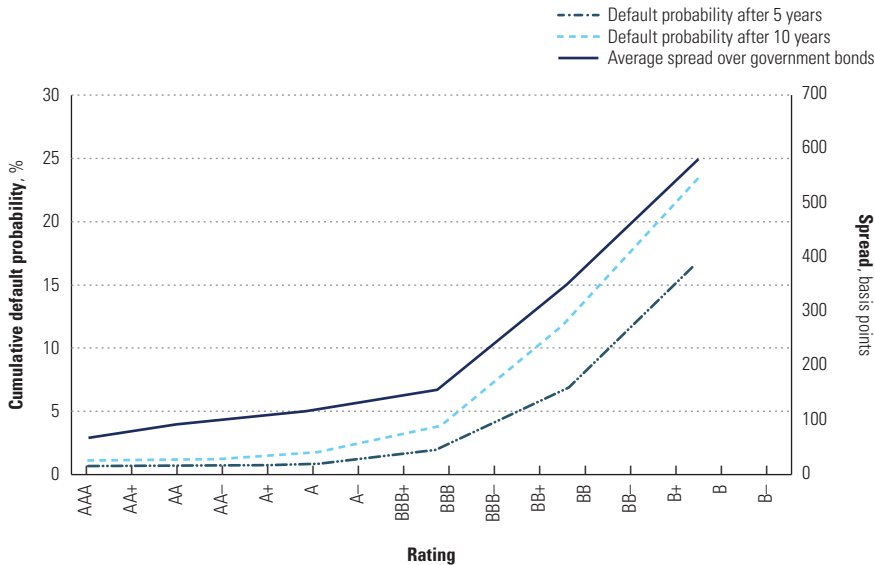
² Median volatility of EBITDA over the prior 5 years in each sector.

Source: S&P Capital IQ; Corporate Performance Analytics by McKinsey.

company credit ratios clustered around industry-specific averages, further indicating that each industry has its own effective capital structure.²⁹

From a company's credit rating, you can also estimate the interest rate payable on its debt funding. The difference between the yields on corporate bonds and risk-free bonds—the credit spread—is greater for companies with lower credit ratings, because their probability of default is higher. Exhibit 33.10 plots cumulative default probabilities against the credit ratings over five and ten years and the average credit spread for each rating. The credit spread reflects the increasing default probability almost proportionally, but for ratings below the investment-grade benchmark of BBB, it increases more sharply. One explanation is that some institutional investors cannot invest in debt that is below investment grade (BBB–), so the debt market is considerably smaller for below-investment-grade debt, and interest rates correspondingly higher.

²⁹ E. Schwarz and R. Aronson, "Some Surrogate Evidence in Support of the Concept of Optimal Financial Structure," *Journal of Finance* 22, no. 1 (1967): 10–18.

EXHIBIT 33.10 **Default Probability and Credit Spread**

Source: S&P Capital IQ; Corporate Performance Analytics by McKinsey.

PAYOUTS TO SHAREHOLDERS

Most successful companies, at some point, find it virtually impossible to reinvest all the cash they generate. In that case, there is little alternative but to return surplus cash to shareholders. Although some executives might consider that a failure to find value-creating investments, it is actually an inevitable consequence for maturing companies with high returns on capital and moderate growth. For example, a company with \$1 billion of net operating profit after taxes (NOPAT), a return on invested capital of 25 percent, and annual revenue growth of 5 percent needs net investments of only \$200 million per year to continue its growth at that rate. That leaves \$800 million of surplus cash flow for additional investments or payouts to shareholders (see Exhibit 33.11). Finding \$800 million of new investment opportunities at attractive returns in every year is a challenge in many industries. Reinvesting all its surplus cash flow in new opportunities at its current return on capital of 25 percent would imply that the company grows revenues by 20 percent each year.

The payout levels for different combinations of return and growth in Exhibit 33.11 indicate that for most successful companies, even those with double-digit growth rates, the implications will eventually be similar: there is no choice but to return substantial amounts of cash to shareholders. Between 2002 and 2014, Procter & Gamble returned \$113 billion in dividends and share repurchases to its shareholders, representing more than 90 percent of

EXHIBIT 33.11 **Surplus Cash Flow, Given Earnings of \$1 Billion**

Surplus under given conditions, \$ million

Projected return on capital, %	Projected growth rate, %		
	15	10	5
50	700	800	900
25	400	600	800
15	—	333	667

its net earnings over these years. Even for a company like Procter & Gamble, it would have been close to impossible to reinvest that amount of cash, given that it had already spent some \$2 billion per year on R&D and \$8 billion on advertising.

Companies with cash surpluses have three basic alternatives for paying out the surpluses to shareholders: dividend increases, share repurchases, and extraordinary dividends. All three provide a positive signal to the capital market about a company's prospects. The potential negative signal that a cash payout could send is that the company has run out of investment opportunities. This assumes that investors did not already know that the company was generating more cash flow than it could reinvest. However, such cases are extremely rare; investors typically anticipate payouts long before managers make that decision, as illustrated by the simple math in our example in Exhibit 33.11.³⁰

Dividends

Companies that increase their dividends receive positive market reactions averaging around 2 percent on the day of announcement.³¹ For companies that initiate dividend payments, the impact is even greater.³² In general, investors interpret dividend increases as good news about the company's long-term

³⁰ One such rare example is that of Merck, one of the largest pharmaceutical companies worldwide. In 2000, it announced a \$10 billion share repurchase, which led to a 15 percent fall in its share price in the next four weeks (although the initial price reaction was favorable). This would have happened if investors assumed that Merck had been unable to find interesting R&D opportunities and could no longer maintain its long-term earnings growth target of 20 percent. See J. Pettit, "Is a Share Buyback Right for Your Company?" *Harvard Business Review* 79, no. 4 (2001): 141–147.

³¹ See, for example, S. Benartzi, R. Michaely, and R. Thaler, "Do Changes in Dividends Signal the Future or the Past?" *Journal of Finance* 52, no. 3 (1997): 1007–1034; and J. Aharony and I. Swarey, "Quarterly Dividends and Earnings Announcements and Stockholders," *Journal of Finance* 35, no. 1 (1980): 1–12.

³² P. Healey and K. Palepu, "Earnings Information Conveyed by Dividend Initiations and Omissions," *Journal of Financial Economics* 21, no. 2 (1988): 149–175.

outlook for future earnings and cash flows. On average, they are right, according to the evidence. Most companies that increase their dividend payout usually do so after strong earnings growth and when they are able to maintain such high levels of earnings in the year following the dividend increase. Companies that start paying dividends for the first time typically continue to experience high rates of earnings growth.

The drawback of increasing dividends is that investors interpret this action as a long-term commitment to higher payouts. Companies, especially in the United States, have created expectations among shareholders that dividends will be cut only in case of severe setbacks. The stock market greatly penalizes companies for cutting dividends from customary long-term levels. Between 1994 and 2008, only 5 percent of U.S. listed companies with revenues greater than \$500 million cut their dividends, and in almost every case, the company faced a severe financial crisis.

A few companies do not commit to dividends or dividend growth rates that are supposed to be upheld even in the face of adverse events or declining business conditions. Instead, they have variable dividend policies and try to manage investor expectations of future payouts by explicitly relating the dividend payouts to business results. For example, in 2016 Anglo-Australian resources companies BHP and Rio Tinto adopted a dividend payout policy in which dividends are more closely related to underlying business results. BHP switched to a minimum dividend payout ratio equal to 50 percent of underlying profit, with additional payouts made in the form of special dividends or share repurchases if and when the company's financial position allowed (for example, to pay out divestment proceeds).³³ From a value creation perspective, a variable dividend policy is not better or worse than a fixed (or progressive) dividend policy. But it does create more financial flexibility, saving managers from feeling compelled to uphold dividends even if that means forgoing attractive investment opportunities or divesting assets.

Managers considering increases in dividend commitments—whether in the form of fixed dividends or a payout ratio for variable dividends—should be confident that future cash flows from operations will be sufficient to pay for capital expenditures as well as higher dividends. Furthermore, a higher dividend payout could lead to higher taxable income for shareholders, depending on the jurisdiction and their individual tax position. Such shareholders could suffer a tax loss if a company would make unexpected, significant changes to the dividend payout ratio. In other words, dividend increases are useful to handle structural cash surpluses over time but much less suitable for a one-time surplus payout.

³³ BHP's new dividend policy was announced in the release of the second half-year results for 2015 on February 23, 2016. Rio Tinto's dividend policy states that it expects to pay out dividends in a range of 40 to 60 percent of aggregate underlying profit through the cycle (<https://www.riotinto.com/investors-87.aspx>).

Share Repurchases

In the early 1980s, share repurchases represented less than 10 percent of cash payouts to shareholders. Since then, they have gained notable importance as an alternative way to distribute cash to shareholders, mainly because key regulatory limits for corporations to purchase their own shares were removed in the United States in 1982.³⁴ By 1999, for example, share repurchases totaled \$181 billion, close to the \$216 billion in regular dividend payments for companies listed on the New York Stock Exchange.³⁵ Even in the wake of the stock market downturn in 2000, major companies in different sectors have continued to repurchase shares on a large scale; examples include ExxonMobil, IBM, Marks & Spencer, Shell, Unilever, and Viacom. In 2018, about 60 percent of cash distributions to shareholders in the United States were share repurchases.

Investors typically interpret share repurchases positively, for several reasons. First, a share buyback shows that managers are confident that future cash flows are strong enough to support future investments and debt commitments. Second, it signals that the company will not spend its excess cash on value-destroying investments. Third, buying back shares indicates to investors that management believes the company's shares are undervalued. If management itself buys back shares, this effect is reinforced. Research shows that because of this signaling, share prices historically increased 2 to 3 percent on average on the day of announcement for smaller repurchase programs (in which less than 10 percent of shares outstanding were acquired through open-market transactions).³⁶ However, these results were mostly driven by share price increases for smaller companies. In addition, repurchases have become a regular payout instrument, so that their signaling effect has declined over the years.

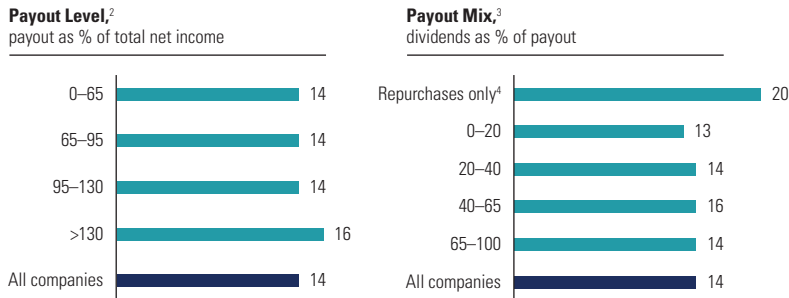
These signaling effects should not be confused with value creation for shareholders, as they only reflect higher market expectations of future performance. If the company does not deliver against these higher expectations, the share price will come down again. As is the case for all cash payouts to shareholders, repurchases do not create value for shareholders, because they do not increase the company's cash flows from operations. This is confirmed by empirical evidence that earnings multiples are not related to the amount or the form of the cash returns, whether in dividends or via share buybacks (see Exhibit 33.12).³⁷

³⁴ Following Rule 10b-18 of the U.S. Securities and Exchange Commission.

³⁵ See Pettit, "Is a Share Buyback Right for Your Company?"

³⁶ In smaller programs, companies typically buy their own shares at no premium or a limited premium in so-called open-market purchases. Larger programs are often organized in the form of tender offers in which companies announce that they will repurchase a particular number of shares at a significant premium. See, for example, R. Comment and J. Jarrell, "The Relative Signaling Power of Dutch-Auction and Fixed Price Self-Tender Offers and Open-Market Repurchases," *Journal of Finance* 46, no. 4 (1991): 1243–1272; and T. Vermaelen, "Common Stock Repurchases and Market Signaling: An Empirical Study," *Journal of Financial Economics* 9, no. 2 (1981): 138–183.

³⁷ See B. Jiang and T. Koller, "Paying Back Your Shareholders," *McKinsey on Finance*, no. 39 (2011): 2–7.

EXHIBIT 33.12 **Valuation Unrelated to Payout Level or Payout Mix**Median enterprise-value-to-EBITDA multiple,¹ end of year 2007¹ Median multiple of nonfinancial companies in S&P 500 index.² Payout defined as dividends paid plus share repurchases, 2002–2007.³ Average proportional share of dividends in total payout, 2002–2007.⁴ This category's higher level results from a higher proportion of fast-growing companies relative to other categories.

Source: Corporate Performance Analytics by McKinsey.

Nevertheless, two myths about share repurchases seem to persist among analysts and managers. The first is that managers can create value by repurchasing shares when they are undervalued.³⁸ Managers have inside information and could be in a better position than investors to assess when the company's shares are undervalued in the stock market and to buy these at the right time. Buying the undervalued shares would create value for those shareholders who hold on to them. However, the empirical evidence shows that companies rarely pick the right time to buy back shares.³⁹ For 2001 through 2010, a majority of the S&P 500 companies bought back shares when prices were high, and few bought shares when prices were low. In fact, the timing of share repurchases by more than three-quarters of S&P 500 companies resulted in lower shareholder returns than a simple strategy of equally distributed repurchases over time would have generated (see Exhibit 33.13).

The second myth is that repurchases create value simply because they increase earnings per share (EPS). The implicit assumption is that the price-to-earnings ratio (P/E) remains constant. As explained in Chapter 3, the logic is flawed: when share repurchases are financed with excess cash or new debt, a company's EPS indeed goes up, simply because the P/E for cash or debt is higher than for the company's equity.⁴⁰ However, after the repurchase, the

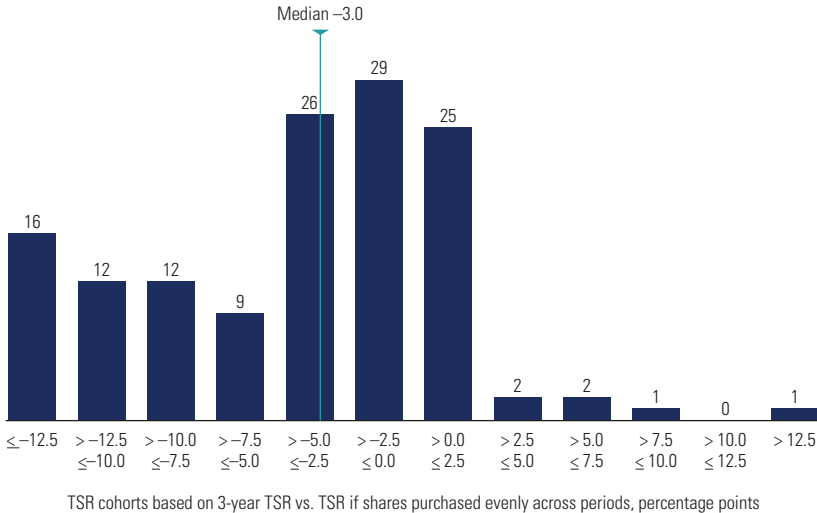
³⁸ See B. Jiang and T. Koller, "The Savvy Executive's Guide to Buying Back Shares," *McKinsey on Finance*, no. 41 (2011): 14–17.

³⁹ Some academic studies have concluded that companies do, in fact, time their repurchases well. Those findings, however, are driven primarily by smaller companies that make a one-time decision to repurchase shares. Once those smaller companies are excluded, the smart-timing effect disappears.

⁴⁰ We define the P/E here in general terms as the market value of an asset or liability divided by its after-tax earnings contribution. The P/Es for cash and debt are the inverse of their after-tax interest rates and are typically higher than for the company's equity.

EXHIBIT 33.13 **Relative Performance of Timing Share Repurchases**

Number of companies per TSR cohort,¹ 2004–2010



¹ Based on 135 S&P 500 companies that repurchased shares from 2004 to 2010. TSR is total shareholder returns.

Source: Corporate Performance Analytics by McKinsey.

equity P/E will be lower because the company's leverage has increased.⁴¹ The increase in EPS does not lead to value creation for shareholders, because it is exactly offset by the decline in P/E. Of course, in a large sample of companies and over long periods of time, there is always an apparent correlation between EPS growth and total shareholder returns (TSR), but that is entirely attributable to revenue growth and return on capital. After controlling for these value drivers, there is no correlation between a company's share repurchase intensity and its shareholder returns.⁴²

The real value creation from share repurchases can only be assessed in comparison with alternative cash deployments, such as business investments, debt repayments, cash holdings, or dividend payments. Contrary to common beliefs, EPS and P/E's provide no guidance in making the assessment. Alternative deployments of cash have a mechanical impact on these metrics that does not necessarily correlate with value creation. This is illustrated in Exhibit 33.14 with a hypothetical company that generates net operating profit after

⁴¹ There are two ways to explain the P/E decline. The first considers the P/E for a company's equity as a weighted average of the P/E's for its operations, cash, and debt. Paying out cash reduces the weight of the relatively high P/E for cash and therefore lowers the P/E for a company's equity. Attracting debt increases the negative weight of its relatively high P/E and lowers the equity P/E as well. The second explanation says that with higher leverage, shareholder risk has increased. This drives up the cost of equity and leads to a lower P/E multiple.

⁴² See O. Ezekoye, T. Koller, and A. Mittal, "How Share Repurchases Boost Earnings without Improving Returns," *McKinsey on Finance*, no. 58 (2016): 15–24.

taxes (NOPAT) of \$100, which translates to an enterprise value of \$1,500 (at an enterprise-value-to-NOPAT multiple of 15 times). The company has an excess-cash position of \$100, no debt, and 100 shares outstanding. It can decide to hold on to the cash or use it to repurchase shares, pay dividends, or invest in operations. Shareholder value increases for the investment alternative because the return on capital exceeds the cost of capital. But it remains unchanged for the other three alternatives, even though the associated changes in EPS or P/E appear to indicate otherwise. The exhibit compares all four alternative cash deployments in detail:

1. *Hold cash.* In this case, the company keeps the excess cash, and net income for the upcoming year is \$102 (assuming the after-tax interest rate on the \$100 cash is 2 percent). The company's value per share is \$16, EPS is \$1.02, and the P/E is 15.7.
2. *Repurchase shares.* The company uses its \$100 in cash to buy back 6.25 units of its own shares (equal to \$100 divided by a share price of \$16). The value per share is unchanged at \$16 (the remaining equity value of \$1,500 divided by 93.75 remaining shares). But the EPS increases to \$1.07, even though no value is created. This is simply due to the fact that the P/E for cash is higher than for shares.⁴³ After the share buyback, the company's equity has a lower P/E because leverage is now higher. The decline in P/E cancels out the increase in EPS, keeping shareholder value unchanged.
3. *Pay dividends.* The company pays a \$1 dividend on each of its 100 shares outstanding. Although the value per share declines from \$16 to \$15, each shareholder still ends up with a total value including dividends of \$16 per share. Again, there is no value creation, but now the EPS declines to \$1.00 because the interest-generating cash has been paid out to the shareholders. The P/E for the company's equity also declines, because leverage increases due to the cash payout. The lower EPS and P/E tie with the decline in value per share of \$1, which is exactly equal to the dividend paid per share.
4. *Invest.* The value for shareholders does change when the company can invest the \$100 in the business at an after-tax return (ROIC) of 15 percent. At a constant enterprise-value multiple of 15 times, the enterprise and equity value will increase to \$1.725 (as NOPAT increases to \$115 from \$100). Because of the high return on investment, the EPS increases to \$1.15, clearly above any other scenario. The value per share is now \$17.25, higher than in all other scenarios, because the business investment creates \$125 additional value for shareholders (\$1.25 per share).

⁴³ The P/E for cash in this example is 50 times (equal to the inverse of the after-tax interest rate of 2 percent).

The erratic pattern of EPS changes across the alternative allocations demonstrates that it does not move in line with value creation. Even though it is highest for the alternative with the highest value creation, this does not mean EPS is a reliable indicator of value creation. For example, assume that the investment would not produce operating earnings in the upcoming year, but only after several years. Of course, the investment would still create value.⁴⁴ But now the EPS would not increase to \$1.15; it would instead decline to \$1.00. Also, changes in the P/E would send the wrong signal. Delayed earnings would further increase the P/E for the investment alternative, not because value creation is higher, but simply because the company's earnings for the upcoming year would be lower. Exhibit 33.14 also shows that share repurchases can destroy value if they prevent the company from pursuing attractive investment opportunities. This underlines that payout decisions, whether repurchases or dividends, always need to be considered as part of a company's overall cash deployment.

EXHIBIT 33.14 **Value Creation from Share Repurchases vs. Alternatives for Cash Deployment**

\$	Hold cash	Repurchase shares	Pay dividends	Invest
Earnings per share				
NOPAT ¹	100.0	100.0	100.0	115.0
After-tax interest income (expense) ²	2.0	—	—	—
Net income	102.0	100.0	100.0	115.0
Number of shares	100.00	93.75	100.00	100.00
Earnings per share	1.02	1.07	1.00	1.15
Enterprise and equity value				
Enterprise value/NOPAT	15	15	15	15
Enterprise value	1,500.0	1,500.0	1,500.0	1,725.0
Cash	100.0	—	—	—
Equity value	1,600.0	1,500.0	1,500.0	1,725.0
Pay out: Dividends			100.0	
Pay out: Share repurchases		100.0		
Equity value including payouts	1,600.0	1,600.0	1,600.0	1,725.0
Value per share				
Value per share	16.00	16.00	15.00	17.25
Dividend per share	—	—	1.00	—
Value per share including dividends	16.00	16.00	16.00	17.25
Price/earnings	15.7	15.0	15.0	15.0

¹ Net operating profit after taxes.

² After-tax interest rate on cash is assumed to be 2% per year.

⁴⁴ Assuming the same 15 percent ROIC and 15 times enterprise value multiple.

When a company then decides to pay out cash to shareholders, there are some good reasons to use share repurchases. In contrast to dividend increases, repurchases offer companies more flexibility in adapting their payouts to unexpected investment needs in a volatile economy. Share buyback programs are not seen as long-term commitments and can be adjusted without influencing investor expectations as much as adjustments to regular dividends would. In addition, they offer investors the flexibility to participate or not. For institutional investors, this means they can choose to uphold the amount invested in a stock—for example, because of a client mandate or because they are tracking an index—without having to reinvest dividends and incur any transaction costs. Finally, share buybacks can result in lower taxes than dividend payments for investors in countries where capital gains are taxed at lower rates. In some countries, individuals have the option to defer taxes on any capital gains and realize such gains in a more tax-efficient manner, potentially years later. Because of their flexibility, share repurchases are a very effective way to pay out any cash surpluses that exceed the level of regular dividends.

Extraordinary Dividends

As an alternative to share repurchases, a company could declare an extraordinary dividend payout, as Microsoft did in 2004 as part of its \$75 billion, four-year cash return program. Microsoft paid out a significant portion in the form of an extraordinary dividend because of its concern that the share repurchase was so massive that it would swamp the liquidity in the market for Microsoft stock. The drawback of extraordinary dividends, compared with share repurchases, is that they offer no flexibility to shareholders and force the cash payout on all of them, regardless of their preferences for capital gains or dividends.

EQUITY FINANCING

If a company is facing a cash deficit and has already reached its long-term leverage target, it has little choice (other than selling noncore businesses, as discussed later in this chapter) but to raise equity or cut its dividends. As with all payout and financing decisions, this does not create or destroy value in itself. But raising equity and—especially—cutting dividends will send negative signals to investors.

As noted, companies are extremely reluctant to cut dividends to free up funds for new investments, because the stock market typically interprets such reductions as a strong signal of lower future cash flows. Share prices on average decline around 9 percent on the day a company announces dividend cuts or omissions.⁴⁵ Furthermore, some investor groups count on dividends being

⁴⁵ Healey and Palepu, “Earnings Information Conveyed by Dividend Initiations and Omissions.”

paid out every year. Skipping these dividends will force these investors to liquidate parts of their portfolios, leading to unnecessary transaction costs. Only very compelling growth opportunities might somewhat mitigate the negative price reactions.⁴⁶ Finally, the amount of funds freed up by cutting dividends is often limited, so dividend cuts alone are unlikely to resolve more substantial funding shortages.

Issuing equity is also likely to lead to a short-term drop in share prices. Typically, share prices decline by around 3 percent on announcements of so-called seasoned equity offerings.⁴⁷ Because investors assume that managers have superior insights into the company's true business and financial outlook, they believe managers will issue equity only if a company's shares are overvalued in the stock market. Therefore, the share price will likely decrease in the short term on the announcement of an equity issuance, even if it is not actually overvalued. A similar price reaction can be expected for various equity-like instruments, such as preferred stock, convertibles, warrants, and more exotic hybrid forms of capital.

DEBT FINANCING

In principle, the amount of debt that needs to be issued or redeemed follows from a company's actual and targeted capital structure. In contrast to equity financing, issuing or redeeming debt typically does not send strong signals to investors about the company's future cash flows.

When issuing debt, companies commit to fixed future interest payments that can be withheld only at considerable cost. Investors also know that debt is more likely to be issued when management perceives a company's share price to be undervalued. As a result, the issuance of debt typically meets with more favorable share price reactions than the issuance of new equity. Empirical evidence shows that the price reaction is typically flat.⁴⁸

Redeeming debt does not meet with significant stock market reactions, either, unless the company is in financial distress. In that case, buying back bonds can send a positive signal to the equity markets. For distressed companies, bond prices go up and down with the enterprise value, just as share prices do. A bond buyback could therefore be a credible signal that management

⁴⁶ L. Lang and R. Litzenberger, "Dividend Announcements: Cash Flow Signaling versus Free Cash Flow Hypothesis," *Journal of Financial Economics* 24, no. 1 (1989): 181–192.

⁴⁷ See, for example, B. Eckbo and R. Masulis, "Seasoned Equity Offerings: A Survey," in *Handbooks in Operations Research and Management Science* 9, ed. R. Jarrow, V. Maksimovic, and W. Ziemba (Amsterdam: Elsevier, 1995); and C. Smith, "Investment Banking and the Capital Acquisition Process," *Journal of Financial Economics* 15, nos. 1/2 (1986): 3–29.

⁴⁸ See, for example, W. Mikkelsen and M. Partch, "Valuation Effects of Security Offerings and the Issuance Process," *Journal of Financial Economics* 15, nos. 1/2 (1986): 31–60; and Smith, "Investment Banking and the Capital Acquisition Process."

believes the bonds are undervalued (and because in this case bonds are similar to equity, this must also mean that shares are undervalued). For example, when the Swiss-Swedish engineering company ABB announced a €775 million bond buyback in July 2004, its share price increased 4 percent on the day of the announcement. The stock market apparently saw the buyback as further evidence that the company was on a trajectory to recover from an earlier financial crisis.

DIVESTITURES OF NONCORE BUSINESSES

As discussed in Chapter 28, companies should regularly monitor whether there are businesses in their portfolio for which they are no longer the best owner. Such businesses could generate more value in the hands of new owners—for example, because of a buyer's distinctive skills, better governance, superior insight and foresight, or strong synergies with their existing businesses. Ideally, portfolio monitoring should form an integral part of a cash deployment process where companies match investment needs across business with funding opportunities from debt, equity financing, and divestitures, also keeping in mind payouts to shareholders.

In recent years, BP, General Electric, and other companies have divested more than \$40 billion in noncore assets, restructuring their corporate portfolios as well as strengthening their balance sheets. Similarly, Royal Philips divested significant parts of its portfolio, such as its lighting business, freeing up cash for investments in organic growth and acquisitions in its core health-care businesses. Such examples underline the importance of always considering divestitures in cash deployment because they form an important source of funds as well as value creation.

CREATING VALUE FROM FINANCIAL ENGINEERING

Managing a company's capital structure with financial instruments beyond straight debt and equity—our definition of financial engineering—typically involves complex and sometimes even exotic instruments such as synthetic leasing, mezzanine finance, securitization, commodity-linked debt, commodity and currency derivatives, and balance sheet insurance. In general, capital markets do a good job of pricing even complex financial instruments, and companies will have difficulty boosting their share prices by accessing so-called cheap funding, no matter how complex the funding structures are. Nevertheless, financial engineering can create shareholder value under specific conditions, both directly (through tax savings or lower costs of funding) and indirectly (for example, by increasing a company's debt capacity so it can raise funds to capture more value-creating investment opportunities). However,

such benefits need to outweigh any potential unintended consequences that inevitably arise with the complexity of financial engineering.

This section considers three of the more common tools of financial engineering: derivative instruments that transfer company risks to third parties, off-balance-sheet financing that detaches funding from the company's credit risk, and hybrid financing that offers new risk/return financing combinations.

Derivative Instruments

With derivative instruments, such as forwards, swaps, and options, a company can transfer particular risks to third parties that can carry these risks at a lower cost. For example, many airlines hedge their fuel costs with derivatives to be less exposed to sudden changes in oil prices. Of course, this does not make airlines immune to prolonged periods of high oil prices, because the derivative positions must be renewed at some point. But derivatives at least give the airlines some time to prepare business measures such as cost cuts or price increases.

Derivatives are not relevant to all companies, and there are many examples where the complexity around the use of derivatives has been badly managed.⁴⁹ In general, derivatives are useful tools for financial managers when risks are clearly identified, derivative contracts are available at reasonable prices because of liquid markets, and the total risk exposures are so large that they could seriously harm a corporation's health.

Off-Balance-Sheet Financing

A wide range of instruments fall under the umbrella of off-balance-sheet financing. These include, for example, real estate investment trusts (REITs), securitization, project finance, synthetic leases, and operating leases. Although the variety of these instruments is huge, they have a common element: companies effectively raise debt funding without carrying all the debt on their own balance sheets. Although they are still referred to as off-balance-sheet financing, new standards for U.S. Generally Accepted Accounting Principles (U.S. GAAP) and International Financial Reporting Standards (IFRS) require that most of these instruments be recognized in the balance sheet, as is also the case since 2019 for operating leases and rentals.

In most cases, off-balance-sheet financing is used to capture tax advantages. For example, many of the largest hotel companies in the United States don't own most of the hotels they operate. Instead, the hotels themselves are owned by other companies, often structured as partnerships or REITs. Unlike corporations, partnerships and REITs don't pay U.S. income taxes; taxes are

⁴⁹ In the 1990s, some high-profile scandals—for example, at Metallgesellschaft and Orange County, California—underlined the need for such caution.

paid only by their owners. Therefore, in the United States, placing hotels in partnerships and REITs eliminates an entire layer of taxation. With ownership and operations separated in this manner, total income taxes are lower, so investors in the ownership and operating companies are better off as a group because their aggregate cash flows are higher.

However, these deals are very complex, because they need to ensure that the interests of the owner and management company are aligned. For example, the deals need to define in advance how the REITs and the hotel companies will make decisions about renovating the hotels, terminating the leases, and other situations where the interests of both parties could conflict. Unfortunately, such potential conflicts are sometimes overlooked or are simply too complex to cover in advance. The owners of Mervyn's (a clothing retail chain in the United States) attempted something similar in 2004 but failed to align the interests of the real estate company and the operating company.⁵⁰ While Mervyn's had plenty of other problems, this structure exacerbated the difficulty of improving the company's performance. Mervyn's filed for bankruptcy in 2008. All its stores were closed and its assets liquidated in 2009.

In other cases, off-balance-sheet financing aims primarily at enabling a company to attract debt funding on terms that would have been impossible to realize for traditional forms of debt. A well-known example is the large-scale securitization of customer receivables undertaken by several auto companies. These companies sold large sums of their receivables to fully owned but legally separate entities.⁵¹ Because the receivables represented relatively sound collateral, these entities had better credit ratings and credit terms than their parent companies. This effectively enabled the companies to tap large sums of debt for investments that otherwise would have been difficult to obtain at similar terms—although one can question whether the investments they made resulted in any value creation, as the securitization structures fell apart in the 2008 credit crisis.

Other successful examples include the use of project financing for building and running large infrastructure projects such as gas pipelines, toll bridges, and tunnels. Companies (or sometimes governments) in emerging markets and with low credit ratings may have difficulty attracting large sums of debt. But they can use project financing to raise cash for the initial investments; once the infrastructure asset is operational, the interest and principal on the debt are paid to the lender directly from the cash flows from the asset's revenues. In this way, the debt service is assured, even if the company itself goes bankrupt.

Some managers find off-balance-sheet financing more attractive because it reduces the amount of assets shown on the balance sheet and increases the

⁵⁰ Emily Thornton, "What Have You Done to My Company?" *BusinessWeek*, December 8, 2008, pp. 40–44.

⁵¹ These represent examples of a so-called special-purpose entity, or—as referred to under U.S. Generally Accepted Accounting Principles—a variable-interest entity.

reported return on assets. That is not a good reason to do it. Investors will see through accounting representations, as discussed in Chapter 7. Furthermore, as already mentioned, following the latest U.S. and international accounting standards, operating leases and special-purpose entities for off-balance-sheet financing need to be fully recognized on the balance sheet.

Hybrid Financing

Hybrid financing involves forms of funding that share some elements of both equity and debt. Examples are convertible debt, convertible preferred stock, and callable perpetual debt. In particular, issuance of convertible debt has seen strong growth over the past decades, and the amount of convertible debt outstanding surpassed €400 billion in 2014.⁵²

Convertible debt, or debt that may be exchanged for common stock in a given proportion within or after a specified period, is an efficient form of debt financing when investors or lenders differ from managers in their assessment of the company's credit risk.⁵³ When the discrepancy is great, it may become difficult or even impossible to achieve agreement on the terms of credit. But a company's credit risk has less impact on credit terms if the debt is convertible. The key reason is that higher credit risk makes the straight-debt component of the convertible less attractive and the warrant component more attractive, so the two components balance each other to an extent. Overall, convertible debt is less sensitive to differences in credit risk assessment and may therefore facilitate agreement on credit terms that are attractive to both parties. This also explains why high-growth companies use this instrument much more than other companies; they usually face more uncertainty about their future credit risk. In 2018, high-tech companies in the United States issued record levels of convertibles, often with so-called call spread overlays that raise the conversion price at which the bond can be exchanged for common equity shares (see Chapter 16 for an example).

Do not issue convertible debt just because it has a low coupon. The coupon is low because the debt also includes a conversion option. It is a fallacy to think that convertible debt is cheap funding. This holds regardless of whether it is straight convertible debt, mandatory convertible debt, convertible debt with or without call spread overlays, or any other of the many variations possible. Also avoid issuing convertible debt simply because it is a way to issue equity against the current share price at some point in the future when share prices will be much higher. That future value is already priced into the conversion options. Furthermore, if the company's share price does not increase sufficiently, the convertible debt will not be converted to equity, and the company will end up with interest-bearing debt instead.

⁵² Bank for International Settlements, *BIS Quarterly Review*, September 2014.

⁵³ See M. Brennan and E. Schwartz, "The Case for Convertibles," *Journal of Applied Corporate Finance* 1, no. 2 (1988): 55–64.

SUMMARY

Although a poorly managed capital structure can lead to financial distress and value destruction, capital structure is not a key value driver. For companies whose leverage is already at reasonable levels, the potential to add value is limited, especially relative to the impact of improvements in returns on invested capital and growth. Managers should refrain from fine-tuning for the optimal capital structure and from simply giving in to any shareholder demands for higher payouts. Instead, they should make sure capital structure and payout decisions are integral parts of a cash deployment that ensures the company has enough financial flexibility to support its strategy while at the same time minimizing the risk of financial distress.

REVIEW QUESTIONS

1. What are the practical guidelines to making the right choices on capital structure and payout?
2. Define optimal capital structure. What is the relationship between optimal capital structure, corporate value, and cost of capital? How does the concept of effective capital structure differ from optimal capital structure?
3. Some companies carry essentially no long-term debt and only a minimal amount of short-term debt in their capital structure. Often these firms are in the technology and biotechnology sectors. Provide an explanation for why well-managed and profitable companies in these sectors appear to under-value the benefits associated with an optimal capital structure.
4. The degree of company financial risk is measured and reported by independent rating agencies such as Standard & Poor's and Moody's. What factors do these rating agencies evaluate when determining a company's financial risk classification? In what range of financial risk classification do we find most companies?
5. Explain why companies with the same credit rating can have very different coverage ratios.
6. Describe the process a manager should employ to establish an effective capital structure and payout policies.
7. Start-up companies typically have little or no debt. Discuss if and how this fits with value maximization, given the cost-benefit trade-offs between different levels of debt and tax savings, overinvestment, business disruption, and investor conflicts.
8. Comment on the following statement: "Since coupon rates on convertible debt are low relative to straight debt, convertible debt is a cheap source of financing."